

$$f(x) = p_0 + p_1 x_1 + p_2 x_2 + \dots$$

$$\sigma(f(x)) = \frac{1}{1 + e^{-f(x)}}$$

$$\sigma = f(x)$$

\* In case of artificial or deep neural network, the activation func<sup>n</sup> → non-linear nature

1) Sigmoid: —  $\sigma(x) = \frac{1}{1 + e^{-x}}$  Range: (0, 1)

Graph: — S-shaped curve  
\* Used in binary classification problems

2) Tanh (Hyperbolic Tangent)  $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$  Range: (-1, 1)

Graph: — S-shaped curve, but symmetric around 0  
\* Inputs have +ve & negative values

3) ReLU (Rectified Linear Unit)  
$$\text{ReLU}(x) = \begin{cases} x & x > 0 \\ 0 & x \leq 0 \end{cases}$$
 Range: [0, ∞)

Graph: — Linear for  $x > 0$ , flat for  $x \leq 0$   
\* Used in deep learning models

4) Leaky ReLU

$$\text{Leaky ReLU}(x) = \begin{cases} x & x > 0 \\ \alpha x & x \leq 0 \end{cases}$$
 Range: (-∞, ∞)  
 $\alpha \rightarrow$  small constant (eg: -0.01)

Graph: — Similar to ReLU but with a small -ve slope for  $x \leq 0$

\* Suitable for models with sparse activations

5) ELU (Exponential Linear Unit)  
$$\text{ELU}(x) = \begin{cases} x & x > 0 \\ \alpha(e^x - 1) & x \leq 0 \end{cases}$$
 Range: (-∞, ∞)  
 $\alpha > 0$  (commonly  $\alpha = 1$ )

Graph: — Exponential decay for  $x \leq 0$  & linear for  $x > 0$

6) Swish: — 
$$\text{Swish}(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}}$$
 Range: (-∞, ∞)



7) Mish :-  $\text{Mish}(x) = x \cdot \tanh(\text{softplus}(x))$   $\text{softplus}(x) = \ln(1+e^x)$   
 Range:  $(-\infty, \infty)$

Sig Derivatives:-

1) Sigmoid:-  $\sigma(x) = \frac{1}{1+e^{-x}}$

$$\begin{aligned}\sigma'(x) &= \frac{d}{dx} \left( \frac{1}{1+e^{-x}} \right) = \frac{0 \cdot (1+e^{-x}) + 1 \cdot (-e^{-x})}{(1+e^{-x})^2} \\ &= \frac{e^{-x}}{(1+e^{-x})^2} = \frac{(1+e^{-x}) - 1}{(1+e^{-x})^2} = \frac{1}{1+e^{-x}} - \frac{1}{1+e^{-x}} \\ &= \sigma(x) - \sigma^2(x) \\ &= \sigma(x)(1-\sigma(x))\end{aligned}$$

2) Tanh Function:-  $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$\begin{aligned}\tanh'(x) &= \frac{e^x(-e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2} \\ &= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2} = 1 - \left( \frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2 \\ &= 1 - \tanh^2(x)\end{aligned}$$

3) ReLU :-  $\text{ReLU}(x) = \begin{cases} x & x > 0 \\ 0 & x \leq 0 \end{cases}$

$$\text{ReLU}'(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

4) Leaky ReLU  $\text{Leaky ReLU}(x) = \begin{cases} x & x > 0 \\ \alpha x & x \leq 0 \end{cases}$

$$\text{Leaky ReLU}'(x) = \begin{cases} 1 & x > 0 \\ \alpha & x \leq 0 \end{cases}$$

5) ELU :-  $\text{ELU}(x) = \begin{cases} x & x > 0 \\ x(e^x - 1) & x \leq 0 \end{cases}$

$$\text{ELU}'(x) = \begin{cases} 1 & x > 0 \\ \alpha e^x & x \leq 0 \end{cases}$$



$$6) \text{ Swish} = \frac{x}{1+e^{-x}} \quad \text{Swish}'(x) = \frac{(1+e^{-x}) - x(e^{-x})}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}} + \frac{x e^{-x}}{(1+e^{-x})^2}$$

$$= \frac{1}{\sigma(x)} + \sigma(x) \left( \frac{x e^{-x}}{1+e^{-x}} \right)$$

$$= \sigma(x) (1 + x e^{-x})$$

$$7) \text{ Mish:} \quad \text{Mish}(x) = x \tanh(\text{softplus}(x))$$

$$\text{Mish}'(x) = \tanh(\text{softplus}(x)) + x \tanh'(\text{softplus}(x)) \cdot \text{softplus}'(x)$$

$$\text{softplus}'(x) = \frac{1}{1+e^{-x}} e^x = \frac{1}{e^{-x}+1} = \sigma(x)$$

$$\tanh'(x) = 1 - \tanh^2(x)$$

$$\text{Mish}'(x) = \tanh(\text{softplus}(x)) + x (1 - \tanh^2(\text{softplus}(x))) \sigma(x)$$

Errors

$\hat{y}_i \rightarrow$  predicted value

$y_i \rightarrow$  actual value

$\bar{y} \rightarrow$  Mean of actual values

1) Absolute error (AE):

$$AE = |y_i - \hat{y}_i|$$

2) Sum of Squared Error (SSE):

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

3) Mean absolute error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

4) Mean squared error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

5) Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

6) Coefficient of Determination ( $R^2$ ):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$



6) Swish:-  $\text{Swish}(x) = x \cdot \sigma(x) = \frac{x}{1+e^{-x}}$

$$\frac{d}{dx} \text{Swish}'(x) = \frac{(1+e^{-x}) - x(e^{-x})}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}} + \frac{x e^{-x}}{(1+e^{-x})^2}$$

$$= \sigma(x) + \sigma(x) \left( \frac{x e^{-x}}{1+e^{-x}} \right)$$

$$= \sigma(x) (1 + x e^{-x})$$

7) Mish:-  $\text{Mish}(x) = x \tanh(\text{softplus}(x))$

$$\text{Mish}'(x) = \tanh(\text{softplus}(x)) + x \tanh'(\text{softplus}(x)) \cdot \text{softplus}'(x)$$

$$\text{softplus}'(x) = \frac{1}{1+e^{-x}} e^x = \frac{1}{e^{-x}+1} = \sigma(x)$$

$$\tanh'(x) = 1 - \tanh^2(x)$$

$$\text{Mish}'(x) = \tanh(\text{softplus}(x)) + x (1 - \tanh^2(\text{softplus}(x))) \sigma(x)$$

### Errors

1) Absolute error (AE):-  $\hat{y}_i \rightarrow \text{predicted value}$   
 $AE = |y_i - \hat{y}_i|$   $y_i \rightarrow \text{actual value}$

2) Sum of Squared Error (SSE):-  $\bar{y} \rightarrow \text{Mean of actual values}$   
 $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$

3) Mean absolute error (MAE):-  
 $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$

4) Mean squared error (MSE):-  
 $MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

5) Root Mean Squared Error (RMSE):-  
 $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$

6) Coefficient of Determination ( $R^2$ ):-  
 $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$



7) Adjusted  $R^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1}$   
 $n \rightarrow$  no. of observations  
 $p \rightarrow$  no. of predictors

8) Binary Cross Entropy (BCE)  $\rightarrow$  Binary class  
 $\hat{y}_i \rightarrow$  predicted probability  
 $y_i \rightarrow$  actual binary label

$$BCE = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

9) Cross Entropy (CE)  $\rightarrow$  Multi-class  
 $y_{ij} \rightarrow$  Binary indicator (1  $\rightarrow$  sample  $i \in J$ , 0 otherwise)  
 $\hat{y}_{ij} \rightarrow$  Predicted probability

$$CE = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^K y_{ij} \log(\hat{y}_{ij})$$

10) Softmax Function  $\rightarrow$  Activation func<sup>n</sup> Converts logits to proba.

$$\text{Softmax}(z)_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}}$$

$z = [z_1, z_2, \dots, z_K]$ : Raw scores for  $K$  classes

11) Kullback - Leibler (KL) Divergence

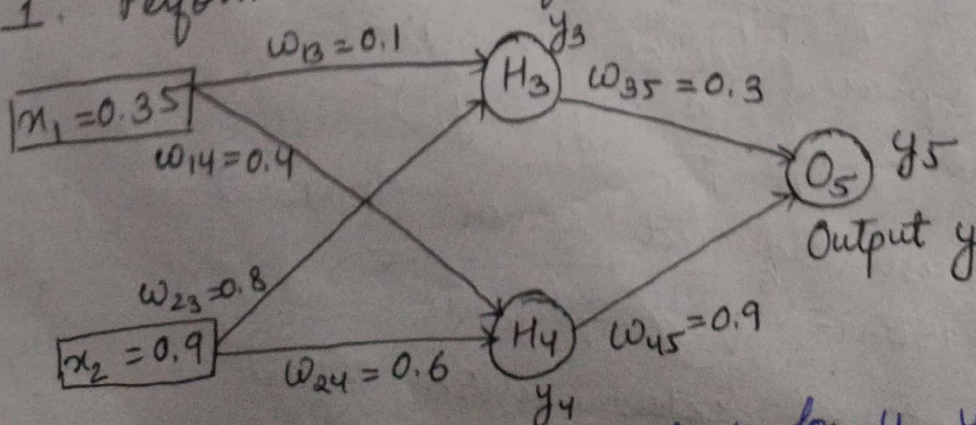
$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

$P \rightarrow$  True probability func<sup>n</sup>  
 $Q \rightarrow$  Predicted probability func<sup>n</sup>

$$D_{KL}(P||Q) = H(P, Q) - H(P)$$

$D_{KL}$ : Always non-negative  
 0  $P=Q$

Q:- Assume that the neurons have a sigmoid activation func<sup>n</sup>, perform a forward pass and a backward pass on the network. Assume that the actual output of  $y$  is 0.5 and learning rate is 1. Perform another forward pass.



Forward Pass: Compute output for  $y_3, y_4, y_5$

$$u_i = (w \cdot x) \quad u_i = F(x_i) = \frac{1}{1 + e^{-u_i}}$$



$$x > -x \quad x > 0 \quad 2 > -2 \quad 2^2 > 2^{-2} \quad 4 > 1$$

$$\log(am) \quad \log(xy) = a \quad \log_2(x) + \log_2(y) = a$$

$$\log_2(xy) = a \quad a \quad m+n \geq a$$

$$2^a = xy$$

$$-0.2231e^x > e^{-x}$$

$$e^x - e^{-x} > 0$$

$$x < -x \quad x < 0 \quad -2 < 2$$

$$e^x < e^{-x}$$

$$e^x - e^{-x} < 0$$

$$e^{2x} - 1 < 0$$

$$e^{2x} < 1$$

$$e^{2x} < e^0$$

$$2x < 0 \quad (x < 0)$$

$$x = (0.8) \frac{1}{1}$$

$$\frac{e^{2x} - 1}{e^{2x} + 1}$$

$$a_1 = (w_{13} * x_1) + (w_{23} * x_2)$$

$$= (0.1 * 0.35) + (0.8 * 0.9)$$

$$= 0.755$$

$$y_3 = f(a_1) = \frac{1}{1 + e^{-0.755}} = 0.68$$

$$a_2 = (w_{14} * x_1) + (w_{24} * x_2)$$

$$= (0.4 * 0.35) + (0.6 * 0.9) = 0.68$$

$$y_4 = f(a_2) = \frac{1}{1 + e^{-0.68}} = 0.6637$$

$$a_3 = (w_{35} * y_3) + (w_{45} * y_4)$$

$$= (0.3 * 0.68) + (0.9 * 0.6637) = 0.801$$

$$y_5 = f(a_3) = \frac{1}{1 + e^{-0.801}} = 0.69$$

$$\text{Error} = y_{\text{target}} - y_5 = 0.5 - 0.69 = -0.19$$

Each weight changed by:

$$\Rightarrow \Delta w_{ji} = \eta \delta_j o_i$$

$$\Rightarrow \delta_j = o_j (1 - o_j) (t_j - o_j) \quad j \rightarrow \text{output unit}$$

$$\Rightarrow \delta_j = o_j (1 - o_j) \sum_k \delta_k w_{kj} \quad j \rightarrow \text{hidden unit}$$

$\eta \rightarrow$  constant  $\rightarrow$  learning rate  
 $t_j \rightarrow$  correct teacher output for unit  $j$   
 $\delta_j \rightarrow$  error measure for unit  $j$

$$\delta_5 = y(1-y)(y_{\text{target}} - y) \rightarrow \text{For output unit}$$

$$= 0.68(1-0.68) * (0.5 - 0.69)$$

$$= -0.0406$$



$$\begin{aligned}
 x &> -x & x > 0 & 2 > -2 & \log(am) & \log(x) + \log(y) = a \\
 -0.2231e^x &> e^{-x} & & 2^2 > 2^{-2} & \log(xy) = a & a \quad m+n=a \\
 x &= 0.8 & e^x - e^{-x} > 0 & 4 > 1 & a^a = xy \\
 x &< -x & x < 0 & -2 < 2 & \\
 e^x &< e^{-x} & & a^{-2} < a^2 & \\
 e^x - e^{-x} &< 0 & \frac{e^{2x} - 1}{e^{2x} + 1} & & \\
 e^{2x} - 1 &< 0 & & & \\
 e^{2x} &< 1 & & & \\
 e^{2x} &< e^0 & & & \\
 2x &< 0 & x < 0 & & \\
 x &= (0.8) \cdot \frac{1}{2} & & & 
 \end{aligned}$$

$$\begin{aligned}
 a_1 &= (w_{13} * x_1) + (w_{23} * x_2) \\
 &= (0.1 * 0.35) + (0.8 * 0.9) \\
 &= 0.755
 \end{aligned}$$

$$y_3 = f(a_1) = \frac{1}{1 + e^{-0.755}} = 0.68$$

$$\begin{aligned}
 a_2 &= (w_{14} * x_1) + (w_{24} * x_2) \\
 &= (0.4 * 0.35) + (0.6 * 0.9) = 0.68
 \end{aligned}$$

$$y_4 = f(a_2) = \frac{1}{1 + e^{-0.68}} = 0.6637$$

$$\begin{aligned}
 a_3 &= (w_{35} * y_3) + (w_{45} * y_4) \\
 &= (0.3 * 0.68) + (0.9 * 0.6637) = 0.801
 \end{aligned}$$

$$y_5 = f(a_3) = \frac{1}{1 + e^{-0.801}} = 0.69$$

$$\text{Error} = y_{\text{target}} - y_5 = 0.5 - 0.69 = -0.19$$

Each weight changed by:

$$\Rightarrow \Delta w_{ji} = \eta \delta_j o_i \quad j \rightarrow \text{output unit}$$

$$\Rightarrow \delta_j = o_j (1 - o_j) (t_j - o_j) \quad j \rightarrow \text{hidden unit}$$

$$\Rightarrow \delta_j = o_j (1 - o_j) \sum_k \delta_k w_{kj}$$

$\eta \rightarrow$  constant  $\rightarrow$  learning rate

$t_j \rightarrow$  correct teacher output for unit  $j$

$\delta_j \rightarrow$  error measure for unit  $j$

$$\begin{aligned}
 \delta_5 &= y(1-y)(y_{\text{target}} - y) \rightarrow \text{For output unit} \\
 &= 0.69(1-0.69)(0.5-0.69) \\
 &= -0.0406
 \end{aligned}$$



For hidden unit:-

$$\begin{aligned}\delta_3 &= y_3(1-y_3)w_{35} * \delta_5 \\ &= 0.68 * (1-0.68) * (0.3 * -0.0406) = -0.00265\end{aligned}$$

$$\begin{aligned}\delta_4 &= y_4(1-y_4)w_{45} * \delta_5 \\ &= 0.6637(1-0.6637) * (0.9 * -0.0406) = -0.0082\end{aligned}$$

$$\Delta w_{45} = n \delta_5 y_4 = 1 * -0.0406 * 0.6637 = -0.0269$$

$$\begin{aligned}w_{45}(\text{new}) &= \Delta w_{45} + w_{45}(\text{old}) \\ &= -0.0269 + 0.9 = 0.8731\end{aligned}$$

$$\begin{aligned}\Delta w_{13} &= n \delta_3 y_1 = 1 * -0.00265 * 0.35 \\ &= -9.275 \times 10^{-4}\end{aligned}$$

$$\begin{aligned}w_{13}(\text{new}) &= \Delta w_{13} + w_{13}(\text{old}) \\ &= -9.275 \times 10^{-4} + 0.1 = 0.0991\end{aligned}$$

$$\Delta w_{14} = n \delta_4 y_1 = 1 * -0.0082 * 0.35 = -2.87 \times 10^{-3}$$

$$\begin{aligned}w_{14}(\text{new}) &= \Delta w_{14} + w_{14}(\text{old}) \\ &= -2.87 \times 10^{-3} + 0.4 = 0.3971\end{aligned}$$

$$\Delta w_{23} = n \delta_3 y_2 = 1 * -0.00265 * 0.9 = -2.385 \times 10^{-3}$$

$$\begin{aligned}w_{23}(\text{new}) &= \Delta w_{23} + w_{23}(\text{old}) \\ &= -2.385 \times 10^{-3} + 0.8 = 0.7976\end{aligned}$$

$$\Delta w_{24} = n \delta_4 y_2 = 1 * -0.0082 * 0.9 = -7.38 \times 10^{-3}$$

$$\begin{aligned}w_{24}(\text{new}) &= \Delta w_{24} + w_{24}(\text{old}) \\ &= -7.38 \times 10^{-3} + 0.6 = 0.5926\end{aligned}$$

$$\begin{aligned}\Delta w_{35} &= n \delta_5 y_3 = 1 * -0.0406 * 0.68 \\ &= -0.0276\end{aligned}$$

$$\begin{aligned}w_{35}(\text{new}) &= \Delta w_{35} + w_{35}(\text{old}) \\ &= -0.0276 + 0.3 = 0.2724\end{aligned}$$



$$a_1 = (w_{13} * x_1) + (w_{23} * x_2) \\ = (0.0991 * 0.35) + (0.7976 * 0.9) = 0.7525$$

$$y_3 = f(a_1) = \frac{1}{1 + e^{-0.7525}} = 0.6797$$

$$a_2 = (w_{14} * x_1) + (w_{24} * x_2) \\ = (0.3971 * 0.35) + (0.5926 * 0.9) = 0.6723$$

$$y_4 = f(a_2) = \frac{1}{1 + e^{-0.6723}} = 0.6620$$

$$a_3 = (w_{35} * y_3) + (w_{45} * y_4) \\ = (0.2724 * 0.6797) + (0.8731 * 0.6620) \\ = 0.7631$$

$$y_5 = f(a_3) = \frac{1}{1 + e^{-0.7631}} = 0.6820$$

$$\text{Error} = y_{\text{target}} - y_5 = 0.5 - 0.6820 \\ = -0.182$$